MIL-T-17442B(SHIPS)
11 May 1964
SUPERSEDING
MIL-T-17442A(SHIPS)
19 March 1953
(See paragraph 6.8)

MILITARY SPECIFICATION

TENSIOMETERS, DIAL INDICATING, HYDRAULIC, REMOTE READING AND MAINTENANCE KIT (MINESWEEPING)

1. SCOPE

1.1 Scope.- This specification covers portable nonmagnetic hydraulic, remote reading tensiometers (used to measure towing tensions during minesweeping operations) and a tensiometer maintenance kit (see 3.7 and 6.1)

1.2 Capacities.- Tensiometers shall be of the following capacities, as specified (see 6 1):

0 - 5,000 pounds tension
0 - 10,000 pounds tension
0 - 20,000 pounds tension

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein

SPECIFICATIONS

MILITARY

MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-P-5516 - Packings and Gaskets, Preformed, Petroleum Hydraulic Fluid Resistant.
MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance.
MIL-I-17214 - Indicator, Permeability; Low-Mu (Go-No-Go).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Service.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

AMERICAN IRON AND STEEL INSTITUTE
(AISI) - Steel Products Manual, Section 24, Stainless Steel.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York 17, New York.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 General requirements.-

3.1.1 Construction - The tensiometer assembly shall consist of a pressure cylinder, an indicating gage and connecting hose. Connecting shackles are not included (see 3.3 2.4). A carrying case (see 3.3.5) shall also be provided. All threaded fittings in the hydraulic system shall be protected against leakage by use of sealing tape such as Teflon or Viton.

3 1.1.1 Screw threads - All screw threads shall be in accordance with Handbook H28.

3.1 1.2 Operating pressure.- The operating pressure of the system shall not exceed 3,000 pounds per square inch (p.s.i.).

3.1.2 Material.- The tensiometer components, carrying case, repair parts, maintenance equipment and maintenance kit box shall have a magnetic permeability not exceeding 2 0 after fabrication (see 4.6.2) unless otherwise specified. With the exception of the crystal, the use of plastic material is prohibited

3.2 Preproduction units.- Two preproduction units shall be furnished by the contractor for the tests specified in 4.5 For the purpose of the tests these units shall consist of a cylinder, gage and hose assembly. They may be fabricated from magnetic material dimensionally identical to the final components (see 4.2).

3 3 Tensiometer components -

3.3.1 Gage -

3.3.1.1 The gage shall be constructed as specified herein (see 6.7), and shall be permanently installed in the carrying case (see 3.3.5). Provisions shall be made for replenishing the gage fluid (3.3.1.10) and hydraulic fluid (3.3.4) with the gage installed.

3.3.1.2 Case.- The gage case shall be made of cast bronze or aluminum and painted in accordance with 3.6.3. Provision shall be made for filling or controlling the level of the gage fluid as specified in 3.3.1.10.

3.3.1.3 Bezel.- The bezel shall be of the same material as the case. It shall be screw type or secured to the case by means of four or more screws inserted from the front.

3.3 1 4 Crystal.- The crystal shall be fabricated from clear acrylic plastic sheet approximately 1/8 inch thick. It shall be heat resistant and of good optical quality and shall be provided with a molded "U" shaped neoprene gasket to provide a seal between the bezel and case.

3.3.1.5 Bourdon tube assembly.- The tube shall be made of an appropriate type of nonmagnetic material such as corrosion resisting steel or beryllium copper. The socket and tip shall be brazed or welded to the tube.

3.3.1.6 Pointer.- The pointer shall be of a plain balanced design, similar to that shown on figure 1, with the indicating end terminating at the inner ends of the minor scale marks The indicating end and tail of the pointer shall be painted white and black respectively, as specified in 3.6.3.

3.3.1.7 Movement assembly.- The movement and linkage assembly shall be of corrosion resisting material. All wearing surfaces shall be of materials suitable to insure minimum friction and wear. The use of plastic parts is not acceptable The movement shall be attached to the socket independent of the case.

3.3.1.8 Dial - The dial shall be made of metal and the visible portion shall be six inches in diameter. It shall be secured to the case by at least three screws. In addition to the scale markings specified herein, the capacity, graduation intervals and manufacturer's name shall be marked on the dial as shown on figures 2 through 4.

3.3 1.8.1 Graduations.- Graduations shall be spaced and numbered as shown on figures 2 through 4. The increment scale marks, and numerals shall be of the dimensions specified in table I.

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Table I - Dimensions of dial markings.

Scale mark	Stroke width inches	Length inches	Height inches
Major	3/32	7/16	---
Intermediate	1/16	5/16	---
Minor	1/32	3/16	---
Numerals[1/]	1/16	---	7/16
Numerals[2/]	1/16	---	7/16
	1/64	---	3/16

1/ 0 - 10,000 pound tensiometer.
2/ 0 - 5,000 and 0 - 20,000 pound tensiometer.

3.3.1.8.2 Finish.- The dial shall have a dull non-reflecting black background permanently marked with white graduations, figures and identification information (see 3.6.3).

3 3.1.9 Pulsation damper.- The gage shall be provided with an adjustable pulsation damper to minimize oscillation of the pointer. A filler plug (an elbow, if necessary) shall be installed at the gage for adding hydraulic fluid to the system from the front without removal of the gage from the case.

3.3 1.10 Gage fluid - The space between the dial and the crystal shall be filled with a light colorless nonflammable oil. An air space of sufficient magnitude shall be left to prevent leakage due to expansion when the gage is subjected to a temperature of 150° Fahrenheit (F.).

3.3.2 Cylinder -

3.3.2.1 The outside diameter and length of the cylindrical section of the tensiometer cylinder assembly shall not exceed the dimensions shown on figure 5. These dimensions shall be the same for the 5,000, 10,000 and 20,000 pound tensiometers.

3.3.2.2 Interchangeability.- In order to insure interchangeability with existing tensiometers it is required that the cylinder body and piston rod (see figure 5) be machined to the dimensions specified in table II and that spanner wrench holes be drilled in the piston as specified in table III. "O" rings shall be of the sizes specified in table IV. In addition, the components required for tensiometers of each capacity, shall be identical with the exception of the gage and the length of hose connecting the gage and cylinder All other like parts shall be completely interchangeable in all respects, including fit and performance

Table II - Cylinder assembly dimensions.

Component	Diameter, inches		Inside diameter, inches		
	Max.	Min	Max.		Min.
Cylinder body	---	---	3.512	or	3.502
			3.252		3.251
Piston rod	1.502	1.492	---		---

Table III - Dimensions of spanner wrench holes

Component	Distance on center, inches	Hole sizes, inches		Number of holes
		Diameter	Depth	
Piston	2-1/4	1/4 drill	1/2	2
	or	or	or	
	2-1/8	9/32 drill	1/4	2

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Table IV - "O" ring dimensions.

AN 6227 dash numbers	Nominal size, inches			Actual molded dimensions inches		Application	Quantity
	I.D.	O.D.	W[1/]	I.D.	W[1/]		
23	1-1/4	1-1/2	1/8	1 234 ± 006	.139 ± 004	Piston dampening (static seal)	1
27	1-1/2	1-3/4	1/8	1 484 ± 006	139 ±.004	Piston rod	1
39	2-7/8	3-1/4	3/16	2.850 ±.015	.210 ±.005	Piston	1
or 41	3-1/8	3-1/2	3/16	3.100 ±.015	.210 ±.005	Piston	1

[1/] W equals diameter of cross section

3 3.2 3 Material.- The cylinder body and head shall be made of an appropriate type of nonmagnetic material such as aluminum bronze or silicon bronze.

3.3.2.4 "O" rings.- "O" rings (see table IV) shall be in accordance with class B of MIL-P-5516.

3.3.2 5 Shackles.- Shackles for use in conjunction with the tensiometers shall not be provided by the contractor (see 6.5).

3.3.2.6 Hose fittings.- A 90 degree fitting shall be provided at the cylinder to eliminate sharp bends in the hose.

3.3.3 Hose.- The hose shall be single wire braid, reinforced, high pressure hydraulic hose manufactured by Electric Hose and Rubber Company, catalog no. 173031-3/16 inch, or equal. It shall be permanently connected to the cylinder and gage by 1/4 inch standard pipe fittings All hose shall have a 3/16 inch internal diameter and shall be provided with connector fittings having 1/4 inch NPT threads. A stainless steel helical spring approximately 3 inches long shall be installed at each end of the hose to prevent excessive bending at the fitting The spring shall be secured to the fitting by means of a groove The length of hose furnished with the various capacity tensiometers shall be as specified in table V.

Table V - Hose lengths.

Tensiometer capacity pounds	Hose length feet
0 - 5,000	12
0 - 10,000	30
0 - 20,000	50

3.3.4 Hydraulic fluid.- The hydraulic fluid used in the system shall be in accordance with MIL-H-5606, or shall be Modil Arrow Hydraulic Oil HFA (Petroleum Base), or equivalent A force feed hand pump designed to prevent aeration and with a ball check to prevent fluid return shall be included in each tensiometer maintenance kit for replacing fluid in the system. A nipple and bushing shall be provided with the pump to facilitate filling the system without removal of the gage from the case

3.3.4.1 Refilling.- Complete instructions shall be provided for replacing the fluid aboard ship (see 3.3.5.6 and 3.7.3.6)

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3.3.5 Carrying case.-

3.3.5.1 The carrying case shall be provided with the unit. The tensiometer gage shall be secured to the back of the interior of the case. Space shall also be provided for stowage of the cylinder, connecting shackles (see 6.5), and hose assembly so that the assembly may be readily removed for use A bracket conforming to the diameter of the cylinder shall be installed under the gage to prevent contact of the cylinder with the gage. The stowage space shall be sufficient to accommodate the 50-foot length of hose without interference with the cover when closed. Carrying cases for all tensiometers shall be identical in size regardless of the tensiometer capacity All sharp corners shall be removed.

3.3.5.2 Material.- The carrying case shall be AISI type 304 corrosion resisting steel, U.S. Gage No. 18. The carrying handle hinge and suitcase type catches shall also be nonmagnetic corrosion resisting steel.

3.3.5.3 Carrying handle.- The carrying case shall be provided with a carrying handle that is permanently attached to the top and located over the center of gravity of the case with all components stowed in their proper place. The handle shall be approximately 5 inches long and hinged to fit flat on the case when not in use

3.3.5.4 Cover.- The entire front of the carrying case shall be covered with a flanged cover hinged from the bottom and secured at the top with 2 suitcase type catches or other suitable means for holding the cover in a closed position

3 3.5 5 Hinge - The hinge shall be piano type full length.

3.3.5 6 Instruction sheet - An illustrated instruction sheet at least 8 x 10 inches in size and with lettering no smaller than 9 point, shall be provided containing instructions for installation, operation, maintenance and refilling of the tensiometer hydraulic system aboard ship. It shall also include the following: "NOTE TO SHIP'S FORCE - Before initial use attach a 1 inch nonmagnetic screw pin anchor shackle to the lugs at each end of the cylinder assembly. Tighten shackle pin and secure with wire seizing " The instruction sheet shall be sealed in plastic (see 3.3.5.6.2) and attached to the inside of the carrying case cover with epoxy or Stabond type cement. When mounted the sheet shall read from the front when the cover is opened at an angle of 90 degrees.

3.3 5.6.1 Printing process - The printing process shall be such that fading of the recorded information will not occur Blueprinting or similar processes are not acceptable

3.3 5 6.2 Sealing.- The instruction sheet shall be laminated between sheets of vinyl chloride plastic The laminating process shall provide a permanent bond between the paper and the plastic

3.4 Accuracy and protection.-

3.4.1 Overload protection.- Surge overloads up to 25 percent shall be sustained without making recalibration necessary. The tensiometer shall be capable of withstanding loads of double its rated capacity without permanent damage.

3 4.2 Accuracy - The error in the reading of the tensiometer gage shall not exceed 1/2 of 1 percent of the rated capacity or 1 percent of the load applied, whichever is the greater.

3.4.3 Zero reading - The design of the gage shall be such that the pointer will automatically return to zero.

3.5 Durability.- The assembled unit shall be sufficiently rugged to withstand rapid vibrations and pulsations in loading.

3.6 Painting.-

3.6 1 All exterior surfaces of the cylinder body and head shall be pretreated with one coat of primer and one coat of semi-gloss light gray equipment enamel If components of the tensiometer assembly such as the gage case (see 3.3.1.2) are fabricated from aluminum, the exterior surfaces shall be anodized prior to priming

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3.6.2 Hydraulic fittings.- Hydraulic fittings such as valves, elbows and plugs at the cylinder and gage shall not be painted.

3.6.3 Gage.- The gage case with the exception of the crystal and bezel shall be anodized (if aluminum), primed and painted gray as specified in 3.6.1. The dial bezel and tail of the pointer (figure 1) shall be painted with non-reflecting black. The indicating end of the pointer, dial graduations, numerals and identification information (figures 2 through 4) shall be painted white. All paint used on the dial and pointer shall be equivalent in oil resistance and general performance to baked epoxy or phenolic, and shall not contain radium or other radioactive materials.

3.6.4 Hose, carrying case and maintenance kit box.- The hose, carrying case and maintenance kit box assemblies shall not be painted.

3.7 Maintenance kit.- Maintenance kits shall be furnished in the quantities specified by the procurement activity. Each kit shall consist of repair parts and maintenance equipment for servicing tensiometers of the capacities specified in 1.2. Each kit shall contain the following parts and equipment compactly packed in a metal box as specified herein:

3.7.1 Repair parts.-

Quantity	Description
1/8 each	"O" rings Nos. 23, 27, 39, 41.
8 each	"O" rings No. 6 (for damper).
1 each	1/4 inch pipe coupling, 1/4 x 1/8 inch pipe bushing and 1/8 inch pipe nipple 5 inches long.
2 each	Hose coupling, 3/16 inch, with 1/4 inch NPT (opposite end grooved for spring).
2 each	Spring for hose.
1/4 each	Dust seal (piston); National Seal No. 40059, or equivalent.
1/2 each	Vent screw 1/4-20 x 3/8 inch RHMS (brass).
2 each	Vent screw 5/16-18 x 3/8 inch RHMS (brass).
1/2 each	Socket set screw, 3/8-16 x 1/2 inch, cone point (CRES).
2 each	Socket set screw, 5/16-18 x 1/2 inch cup point (CRES).
1/8 each	Lock-O-Seal type washer for 1/4 inch vent screw.
8 each	Lock-O-Seal type washer for 5/16 inch vent screw.

1/See figure 5.

3.7.2 Maintenance equipment.-

Quantity	Description
1 set	Wrenches, hexagon key, 1/8, 5/32, and 3/16 inch sizes (Allen type for socket set screws).
1 each	Wrench, damper.
1 each	Wrench, socket end for 1/8 inch pipe plug.
1 each	Wrench, spanner (2 pins, 2-1/8 inch distance on center) (see table III).
1 each	Wrench, spanner (2 pins, 2-1/4 inch distance on center) (see table III).
1 each	Hand pump, force feed with ball check.
2 quarts	Hydraulic fluid (1 quart containers)

3.7.3 Maintenance kit box.-

3.7.3.1 The maintenance kit box shall be of sufficient size for stowing the repair parts and maintenance equipment specified in 3.7.1 and 3.7.2. A removable tray about one inch deep equal in area to that of the inside of the box shall be supported on angles so positioned that the top of the tray will be level with the top of the box. All sharp corners shall be removed.

3 7.3.2 Cover - The top of the box shall have a flanged cover extending over its full length and width. It shall be hinged at the back and secured at the front with a suitcase type catch.

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3.7.3.3 Carrying handle.- The box shall be provided with a carrying handle permanently attached to the cover at the center. The handle shall be identical to that specified in 3.3.5.3.

3.7.3.4 Hinge.- The hinge shall be piano type full length.

3.7.3.5 Material - The box and tray shall be fabricated from AISI type 304 corrosion resisting steel of U.S. Gage No. 18. The carrying handle, hinge, and catch shall also be nonmagnetic corrosion resisting steel.

3.7.3.6 Instruction sheet.- An instruction sheet giving instruction for replenishing the fluid in the system and for cylinder and hose repair shall be attached to the inside of the cover by means of cement such as epoxy or Stabond. This sheet shall also include a repair parts list containing reference numbers, contractor's part numbers, quantities and descriptions of the parts contained in the kit, illustrations showing the location of the parts in the assembly and the steps employed for hose repair. The sheet and lettering thereon shall be as large as possible.

3.7.3.6.1 Printing process.- The printing process shall be as specified in 3.3.5.6.1.

3.7.3.6.2 Sealing.- The instruction sheet shall be sealed in plastic as specified in 3.3.5.6.2.

3.7.3.7 Identification plate.- An identification plate shall be provided and attached to the box as specified in 3.8.1. It shall be marked as specified in 3.8.2.3.

3.8 Marking.-

3.8.1 Brass or aluminum identification plates (rectangular) shall be permanently attached, one each to the cylinder body, carrying case and maintenance kit box with nonmagnetic drive screws or rivets. The plate for the carrying case and for the maintenance kit box shall be attached to the top and cover respectively adjacent to the handle. The cylinder body plate shall be formed to the contour of the body and shall be mounted so as not to adversely affect the strength or operation of the cylinder or the readability of the plate. The plates and markings shall withstand corrosion by weathering and action of salt spray. The size of the letters, numerals and other characters shall be as necessary for easy readability. Identification data shall be applied directly to the gage dial (see 3.3.1.8).

3.8.2 Data.- The identification plates shall contain the data specified in 3.8.2.1, 3.8.2.2 and 3.8.2.3.

3.8.2.1 Cylinder.-

TENSIOMETER CYLINDER, NM LBS. CAPACITY
TENSIOMETER STOCK NO.
MANUFACTURER'S NAME OR TRADEMARK
CONTRACT NO. DATE OF MANUFACTURE
INSPECTED
NAVSHIPS NO. 250-530.

3.8.2.2 Carrying case.-

TENSIOMETER, NM LBS. CAPACITY
TENSIOMETER STOCK NO.
MANUFACTURER'S NAME OR TRADEMARK
CONTRACT NO. DATE OF MANUFACTURE
INSPECTED
NAVSHIPS NO. 250-530.

3.8.2.3 Maintenance kit box -

MAINTENANCE KIT FOR TENSIOMETERS, NM
MAINTENANCE KIT STOCK NO.
MANUFACTURER'S NAME OR TRADEMARK
CONTRACT NO. DATE OF MANUFACTURE
INSPECTED
NAVSHIPS NO 250-530

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3 8.2.4 Gage - An identification plate is not required. All data shall be painted on the dial as specified in 3 3 1.8.

3 9 Technical manuals - Technical manuals (manuscript only), including reproducible artwork shall be prepared containing information necessary for the operation and maintenance of the tensiometer (see 6 6) If desired, the manuscript may be in typewritten form double spaced Parts identification shall be included containing brief descriptive information for basic and repair parts to facilitate identification of the parts in the assembly, and to permit ordering of replacement parts Graphics used to supplement the descriptive matter contained in the text and for parts identification shall consist of photographs or line drawings Color shall not be employed The text of the manuscript shall contain a general description of the tensiometer assembly, detailed descriptions of the components, safety precautions to be observed, a theory of operation, operating and test procedures Preventive maintenance information, checking procedures and corrective maintenance information, such as, troubleshooting, removal, inspection, and replacement of components shall also be included

3 9.1 Preliminary manuscript.- The contractor shall forward two copies of the manuscript in preliminary form, together with two copies of each photograph or line drawing required to illustrate the text, to the procuring activity for approval They shall be submitted by a letter of transmittal

3.9.2 Final manuscript.- Upon approval of the preliminary manuscript three copies of the final manuscript shall be forwarded to the procurement activity for final approval The final manuscript shall be similar to the preliminary manuscript except that all changes to the contents made by the procuring activity shall be incorporated therein Three copies of each photograph or line drawing used to illustrate the text shall be furnished with the final manuscript. Insofar as possible the illustrations shall not exceed 8 x 10 inches After approval, copies of the final manuscript and illustrations shall be forwarded by the procuring activity, one each, to the Bureau of Ships and the U.S Navy Mine Defense Laboratory, Panama City, Florida.

3 10 Drawings - Drawings for the tensiometer and associated equipment are required They shall be in accordance with class A of MIL-D-963

3.10 1 Prints for approval - Two sets of preliminary shop or manufacturing type drawings, in accordance with MIL-D-963, shall be forwarded to the procuring activity as specified in MIL-D-963.

3 10.2 Final drawings.- Two sets of black line, white background prints shall be furnished one each to the Bureau of Ships, and to the procuring activity. The prints shall be direct reading and satisfactory for microfilming in accordance with MIL-D-963

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4 2 Preproduction inspection - The two preproduction units as required by 3 2 shall be subjected to the tests specified in 4 5 Upon completion of the preproduction tests these units shall not be submitted as production equipment

4 3 Sampling -

4 3 1 Lot.- For purposes of sampling, a lot shall consist of all tensiometers of one capacity offered for delivery to one destination

4.3.2 Sampling procedure for quality conformance tests - The number of representative sample tensiometers set forth in table VI shall be selected from each lot for the tests specified in 4 6

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Table VI - Sampling for quality conformance tests.

Number of tensiometers in lot	Number of tensiometers to be selected
1 to 5	All
6 to 10	6
11 to 15	7
16 to 25	8
26 to 40	10
41 to 65	13
66 to 110	18
111 to 180	23
181 to 300	31

4.4 Examination.- Each sample tensiometer shall be examined to determine compliance with the requirements of this specification, in regard to the use of proper materials, accurate fit of parts, dimensions, painting, marking and any other requirements covered by the specification which do not require tests.

4 5 Preproduction tests.- The two preproduction units specified in 3.2 shall be tested in accordance with 4.5.1 and 4 6.1.

4 5 1 Pulsating load and overload.- In order to determine compliance with the requirements for durability under pulsating loads as specified in 3.5, the tensiometer shall be subjected to a pulsating tension test of 15,000 cycles. Each cycle shall consist of increasing the load from 20 percent of rated capacity to 80 percent of rated capacity in 6 seconds and reducing the load to 20 percent of rated capacity in 6 seconds The pulsation damper shall be fully open during this test The test may be continuous or may be conducted in 8 hour intervals. In order to determine conformance with the overload requirement specified in 3.4.1 the load shall be increased from full load to 100 percent overload in 6 seconds At the end of the tests the error in gage reading shall not exceed twice that permitted by 3 4 2 and when recalibrated shall not exceed the error specified.

4.6 Quality conformance tests.- Each sample tensiometer selected in accordance with 4.3 2 shall be tested to determine compliance with the requirements of 4.6 1 and 4.6.2. If any tensiometer is found to be not in conformance with this specification, this shall be cause for rejection of the lot.

4.6.1 Surge overload protection and accuracy.- The tensiometer shall be subjected to a 25 percent surge overload, after which it shall be checked at two or more positions on the gage dial to determine compliance with the requirements for surge overload protection and accuracy as specified in 3 4.1 and 3.4.2, respectively.

4 6.2 Permeability.- The permeability of the tensiometer components, carrying case, repair parts, maintenance equipment (except Allen type set screw wrenches) and maintenance kit box shall be checked with a permeability indicator conforming to MIL-I-17214. The hydraulic hose, connecting the cylinder and gage, has a magnetic wire braid reinforcement and hence is not required to meet this test.

5 PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking - Preservation, packaging, packing and marking shall be as specified in the contract or order (see 6.1).

5.2 Checking hydraulic system - Prior to delivery the contractor shall check the fluid in the hydraulic system of each tensiometer to insure that the tensiometer is in operating condition.

6. NOTES

6.1 Ordering data - Procurement documents should specify the following.

(a) Title, number and date of this specification
(b) Tensiometer capacities required (see 1.2).
(c) Quantity of maintenance kits (see 3 7 and 6.3).

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(d) Information regarding technical manuals and drawings (see 3.9 and 3.10).
(e) Preservation, packaging, packing and marking required (see 5.1).

6.2 Intended use.- The tensiometer assemblies are used for measuring towing tensions during mine sweeping operations.

6.3 Repair parts.- Sufficient repair parts and maintenance equipment are included in the maintenance kit to service each tensiometer specified herein at least four times regardless of capacity. Additional repair parts and equipment are also included in the kit for servicing similar tensiometers now in use

6 4 Stock numbers - The procuring activity will assign separate stock numbers, one each as applicable, to the 0 - 5,000 pound, 0 - 10,000 pound and 0 - 20,000 pound tensiometer assemblies. (The tensiometer assembly consists of cylinder, gage, connecting hose and carrying case.) An additional stock number shall be assigned to the maintenance kit.

6.5 Shackles - Shackles are not contractor furnished. However, stowage for two connecting shackles (when attached one each to the cylinder head and piston rod) shall be provided in the carrying case (see 3.3.5). Table VII is provided for use during fabrication of the piston rod and cylinder head lugs (figure 5) and to assist in determining the stowage space required for the cylinder shackles.

Table VII - Dimensions of cylinder shackles (approx.).

No. req'd	Shackle size inches	Dia. of pin inches	Width between eyes, nom. inches	O.D of eye, max inches	Longitudinal length center of pin to outside of bow, inches	Max. width shackle and pin, inches
2	1	1-1/8	1-11/16	2-5/8	5-1/2	5

6.6 Commercial instruction books - Manufacturer's instruction books may be substituted for the manuscript, provided the text includes (or is modified to include) the information specified in 3 9 Line drawings or photographs shall be in addition to the illustrations included in the manufacturer's instruction books.

6.7 Gages.- Hydraulic pressure gages manufactured by the U.S Gauge Division of Amtek, Inc., under part numbers 36640 or 36641, or equivalent, are generally similar in construction to those specified in 3.3.1.

6.8 CHANGES FROM PREVIOUS ISSUE.- THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - Ships
(Project 6635-N014Sh)

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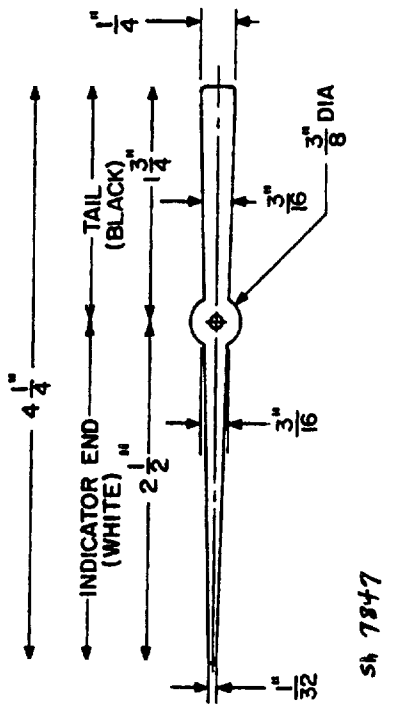


Figure 1 - Pointer.

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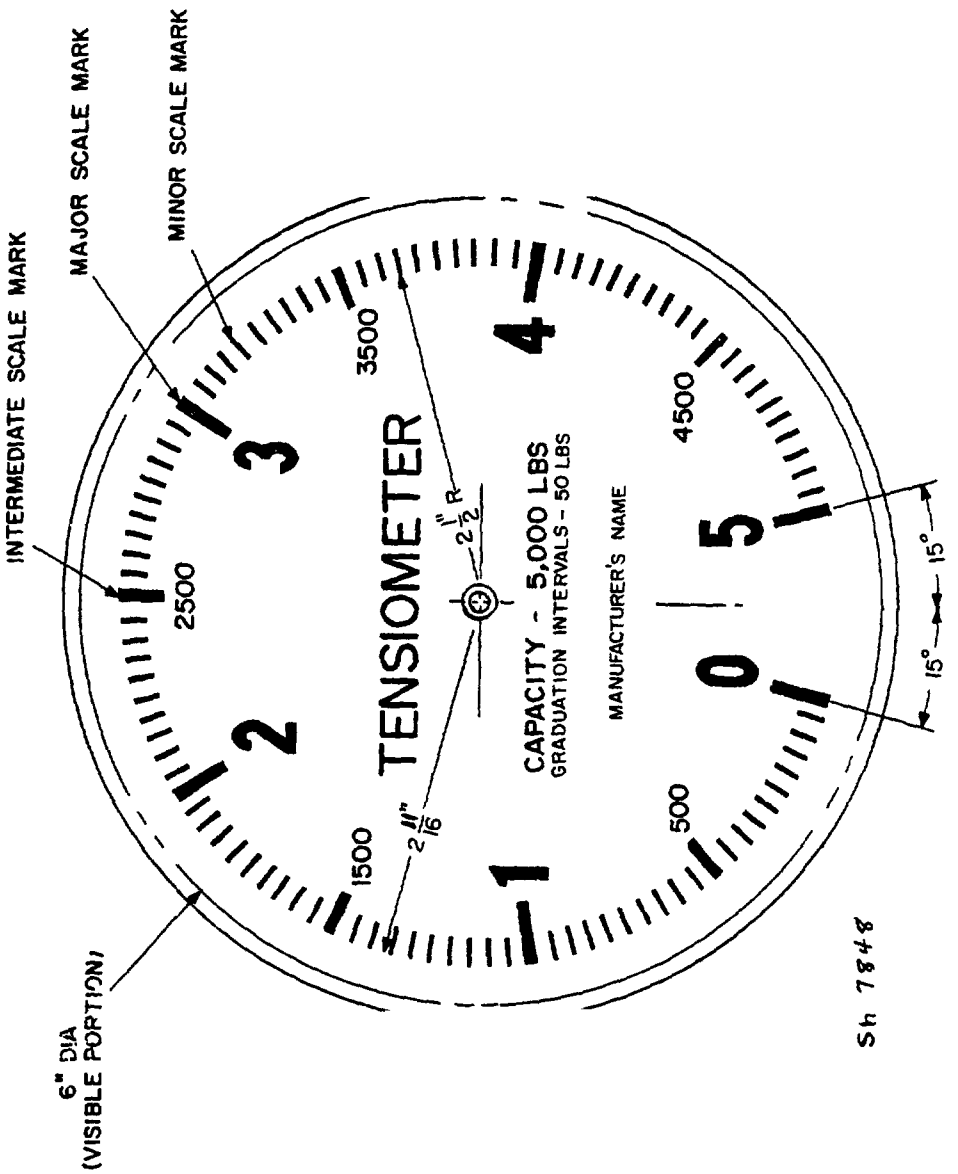


Figure 2 - Gage dial, 0 - 5,000 lb tensiometer.

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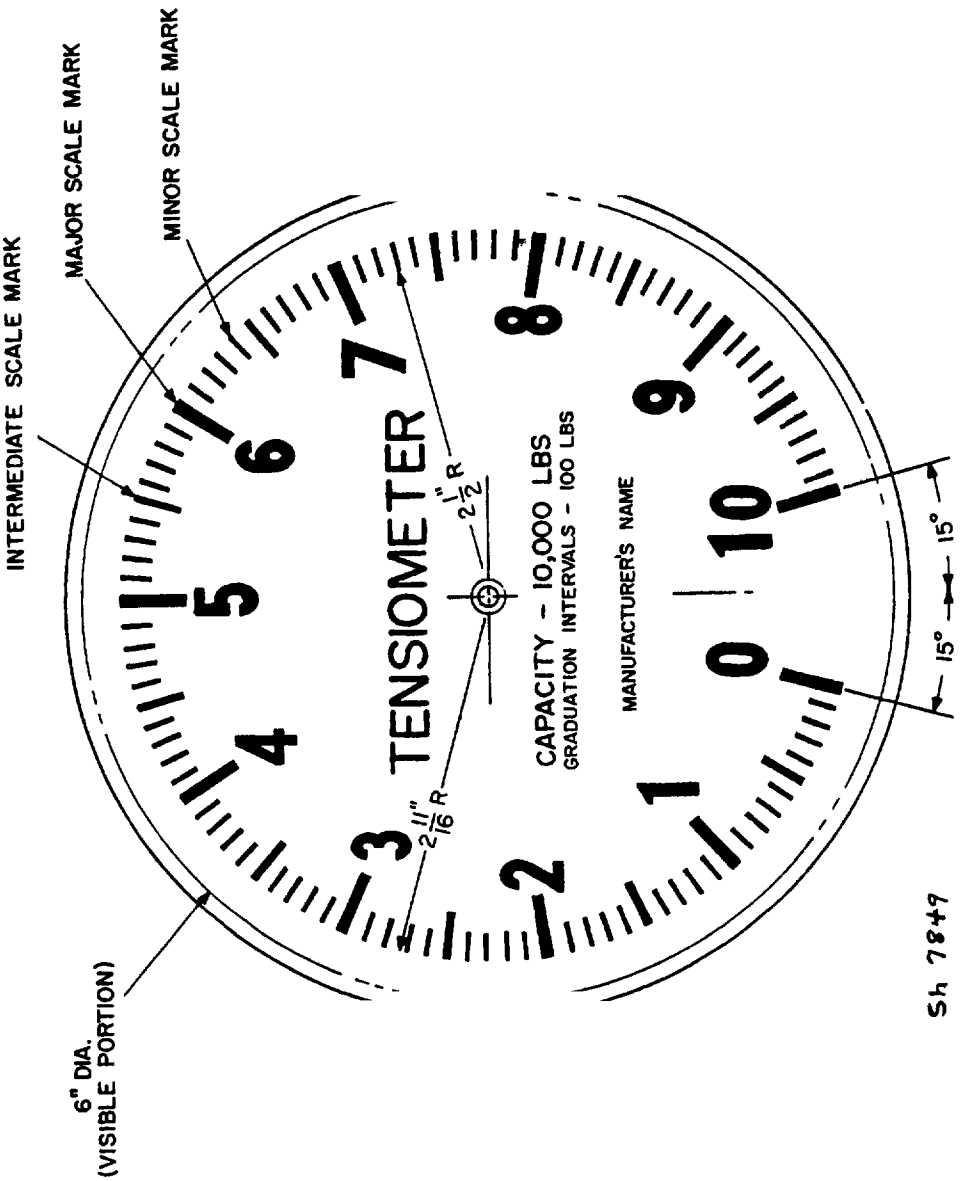


Figure 3 - Gage dial, 0 - 10,000 lb tensiometer.

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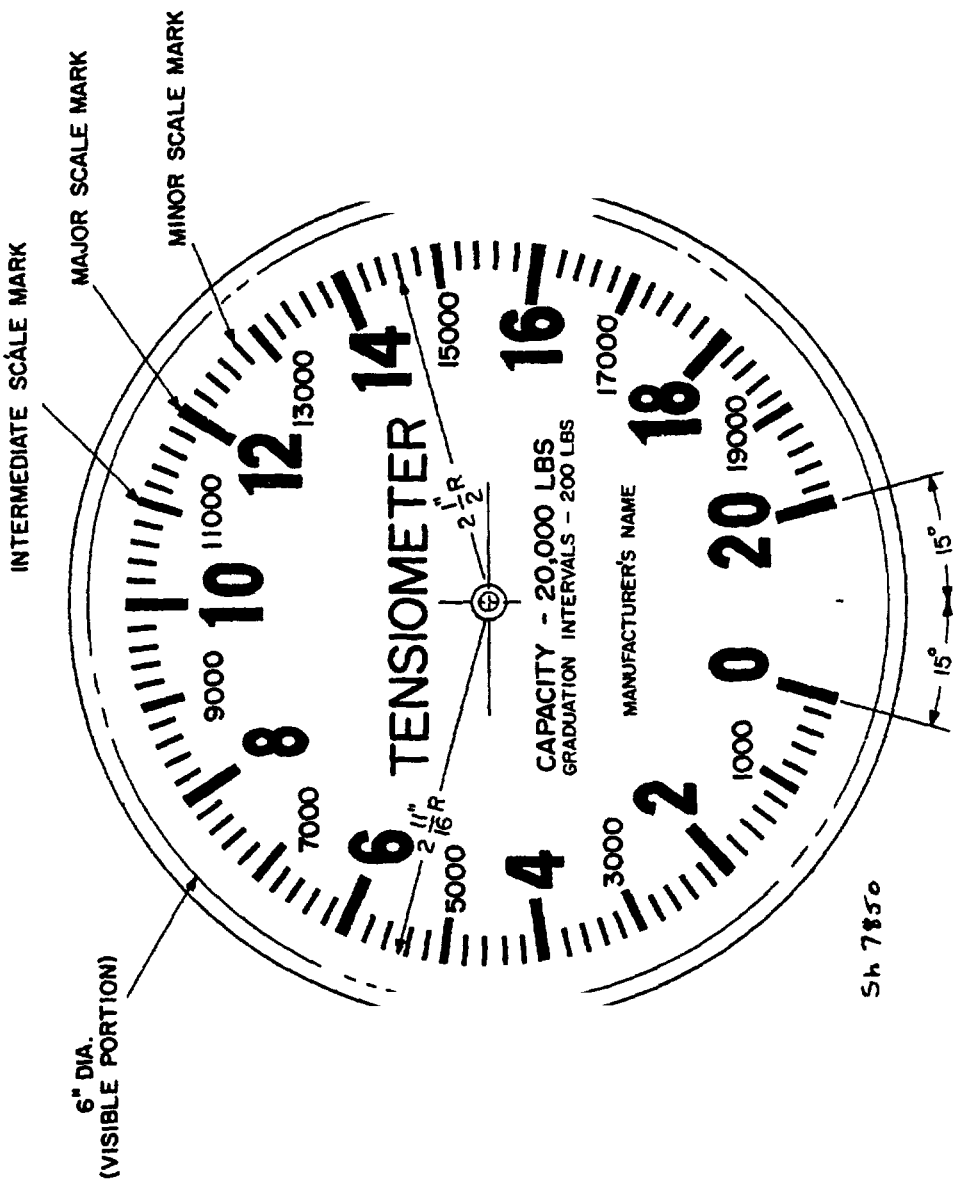


Figure 4 - Gage dial, 0 - 20,000 lb tensiometer.

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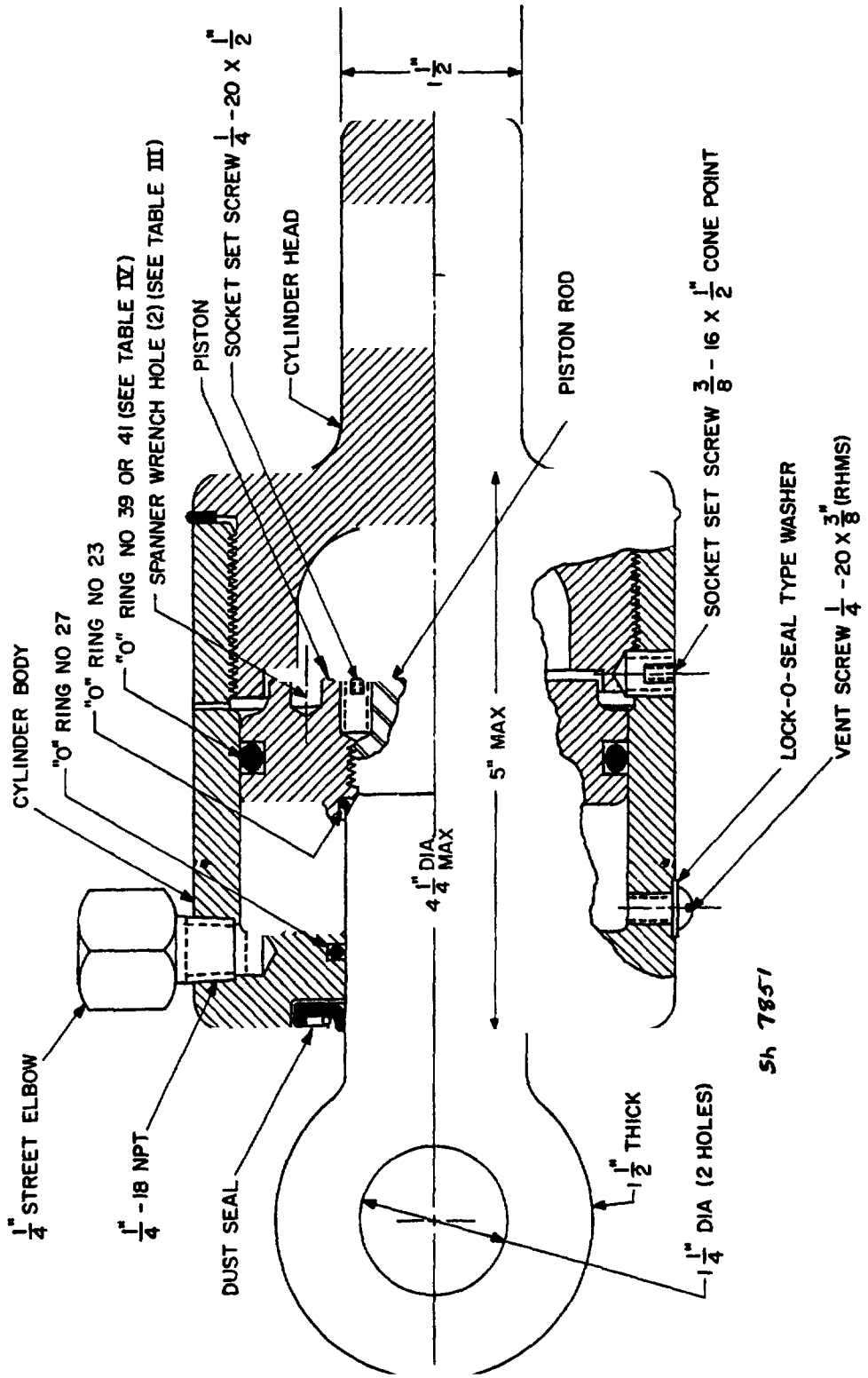


Figure 5 - Cylinder assembly.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 68 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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